Defn A permutation of a set P is a bijective function

$$f: P \to P.$$

In other words, f has the following property:

$$\boxed{\text{for each } y \in P, \text{ there exists a unique } x \in P \text{ with } f(x) = y.}$$

Notation: $x = f^{-1}(y)$, $f^{-1}: P \to P$ is the inverse of f

We can use permutations to build a cryptosystem, called a

substitution cipher:

- plaintexts = ciphertexts = elements of P, an arbitrary (finite) set
- Key space = set of all permutations of P
- Encryption: $e_f(x) = f(x)$, where f is a permutation of P and $x \in P$.
- Decryption: $d_f(y) = f^{-1}(y)$

Exercise: $d_f(e_f(x)) = x$ for all $x \in P$.

Hence: this really is a cryptosystem.

Example $P := \{A, \dots, Z\}$. Define a key (= permutation of P) by

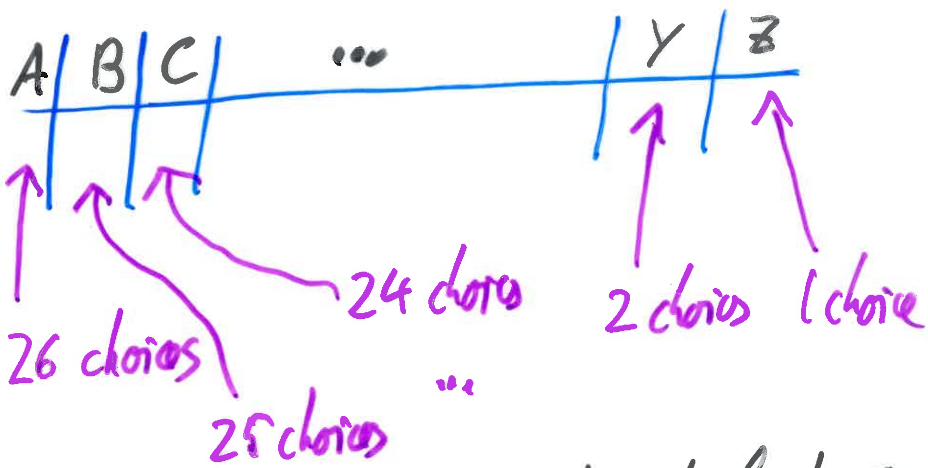
A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
K	V	I	R	U	X	P	Q	B	Y	S	J	T	M	W	C	E	H	L	A	D	O	Z	G	F	N

Encryption: MESSAGE $\rightsquigarrow$ TULLKPU

Decryption: LUIHUA $\rightsquigarrow$ SECRET

Q: Is this more secure than shift ciphers?

How many possible keys are there? Let's build (all) permutations:



Total: $26! \approx 4 \cdot 10^{26}$ keys

If we assume that it takes at least one nanosecond ($= 10^{-9}$s) to try out one key, then a brute force search through the entire key space might take $\geq 1.28 \cdot 10^{10}$ years ($\approx$ 13 billion years).

Current estimate for the age of the universe: $\approx$ 13.8 billion years

Hence: A naive brute force search of the key space is infeasible for substitution ciphers

Q: Does that mean that they are secure?

Breaking substitution ciphers: letter frequency analysis

The fundamental weakness of substitution ciphers is that every plaintext symbol is always mapped to the same ciphertext symbol (for a fixed, given key).

Hence: statistical properties of letters in the original message remain completely intact after encryption.

It turns out that natural languages have distinct distribution patterns of letters in "common" texts.

English:

Letter	A	B	C	D	E	...
Frequency (approx.)	8.16%	1.49%	2.78%	4.25%	12.7%	...

The most common letters (decreasing order of frequency): ETAOIN...

More refined statistics incorporate spaces (more frequent than E!), punctuation symbols, etc.

Frequency tables for pairs, triples, ... of characters also exist.

In practice, such tables can usually be used to decrypt a message produced using a substitution cipher... by recovering the key.

Breaking cryptosystems: common attack vectors

- Mathematics: find and exploit weaknesses of the method
- Side-channel attacks: decrypt a message using information related to the practical implementation of a cryptosystem

Ex: Monitor the power consumption during encryption to learn something about the key being used.

- Social engineering attacks: e.g. trick or coerce people into revealing keys

Punch line: We need good cryptosystems, good implementations, and security awareness.